

Single-Shunt FOC Guide

Power Application Controller[®]





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1 INTRODUCTION

Single shunt operation can be advantageous in cost savings due to the fact that only one current-sensing shunt is necessary compared to the 3-shunt configuration. This cost advantage is mostly appreciated in applications which use high phase currents, typically low voltage applications, where due to the high currents involved require shunt resistors with higher power ratings. High voltage applications typically use lower currents and therefore the shunt resistors require low power ratings.

The best candidates for single shunt operation are applications with the following characteristics:

- 1) Slow acceleration and not very dynamic.
- 2) Operate mostly in the higher speed range of the motor.
- 3) Customer is not concerned about the additional audible noise that could result from 1-shunt operation during startup and low speed range operation.
- 4) Customer does not need to achieve absolute maximum motor speed.

Some examples of good candidates for 1-shunt operation are low voltage vacuum cleaners and some fans.

2 BASICS OF SINGLE-SHUNT OPERATION

2.1 Phase Current Sampling and Reconstruction

The basic idea behind single-shunt current measurement is that current flows through the shunt only when 2 phases are either in HI state or in LO state. If all 3 phases are in HI state or in LO state then current does not flow through the shunt since current will be re-circulating through the power switches.

Figure 1 below shows the sampling points where the single-shunt firmware samples the shunt current while Figure 2 illustrates what the voltage across the shunt may look like for the same duty cycles represented in Figure 1. The 2 sampling points are described below:

- 1) Sampling Point #1 occurs just before the falling edge of the phase with the middle duty cycle. At this point, the lowest duty phase is already in LO state and the highest duty phase is still in HI state. Since 2 phases are in HI state at this point, current is flowing through the shunt. The relationship between the value of the current flowing through the shunt and the actual phase currents at this point is dictated by the following equation:

$$I_{\text{shunt}} = -I_{\text{lowestDC}} = I_{\text{middleDC}} + I_{\text{highestDC}}$$

- 2) Sampling Point #2 occurs just before the falling edge of the phase with the highest duty cycle. At this point, the lowest and middle duty phases are already in LO state and the highest duty phase is still in HI state. Since 2 phases are in LO state at this point, current is flowing through the shunt. The relationship between the value of the current flowing through the shunt and the actual phase currents at this point is dictated by the following equation:

$$I_{\text{shunt}} = I_{\text{highestDC}} = -(I_{\text{middleDC}} + I_{\text{lowestDC}})$$

Because $I_{\text{lowestDC}} + I_{\text{middleDC}} + I_{\text{highestDC}} = 0$, by having the values for I_{lowestDC} and for $I_{\text{highestDC}}$, then I_{middleDC} can always be calculated.

Note that each of the 3 phases cycle through being the lowest, middle, and highest duty cycles at different points throughout the electrical revolution. This is taken into account when doing the current reconstruction of the 3 phases.

Also note that the sampling is done just before the PWM edge in order to make sure that the sample is taken before the edge generates any noise than can corrupt the current sample.

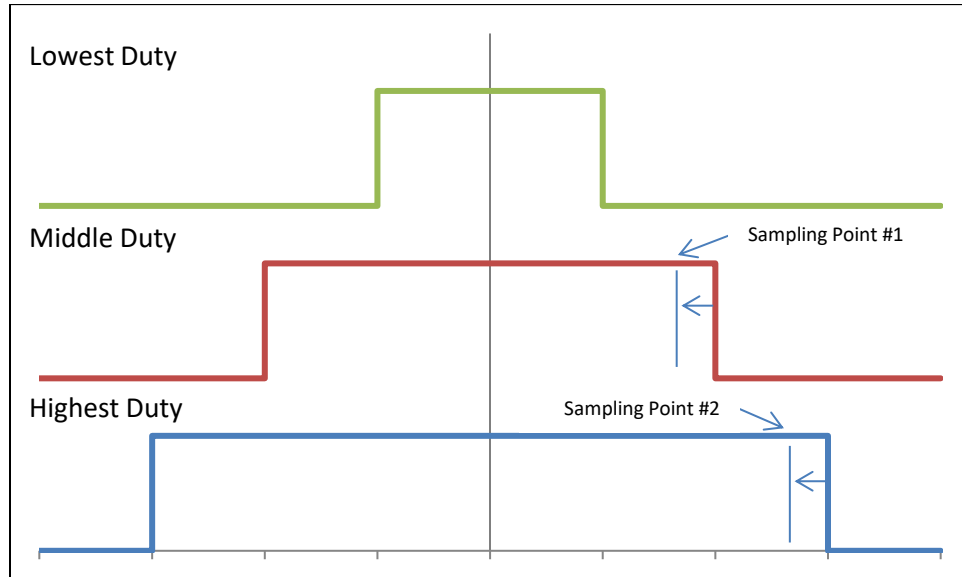


Figure 1: 1-Shunt current sampling points

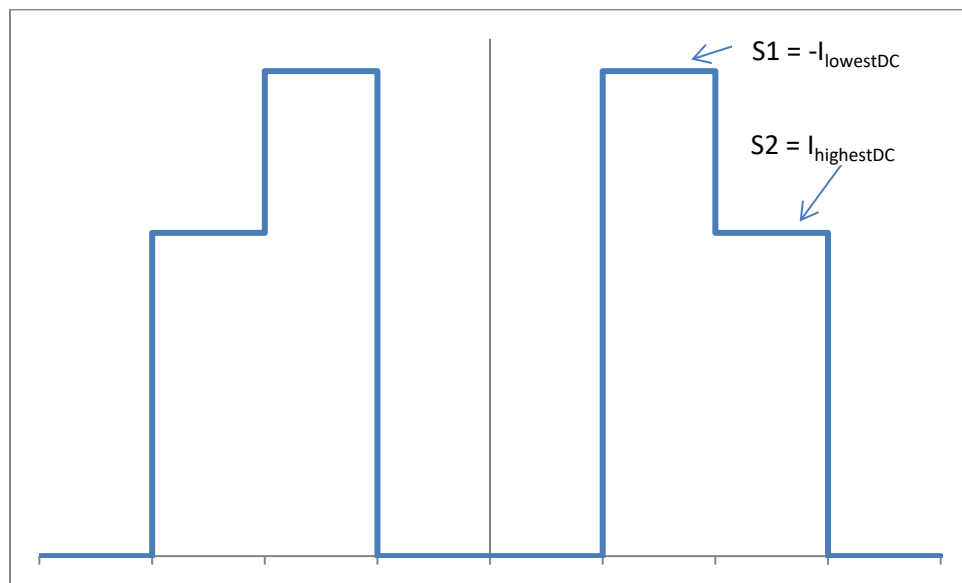


Figure 2: Resultant voltage waveform across shunt for Figure 1

2.2 Minimum Gap Forcing

By studying Figure 1 and Figure 2, it was concluded that the current sampling can only occur at specific points where the current is measurable. By further analysis though, it becomes evident that as the duty cycles change to represent the rotating drive voltage vector, the resulting duty cycles may not satisfy these requirements and therefore the current will not be measurable. Figure 3 and Figure 4 illustrate one such case where the gap between the PWM edges is too small to be able to complete an ADC conversion before the next edge occurs.

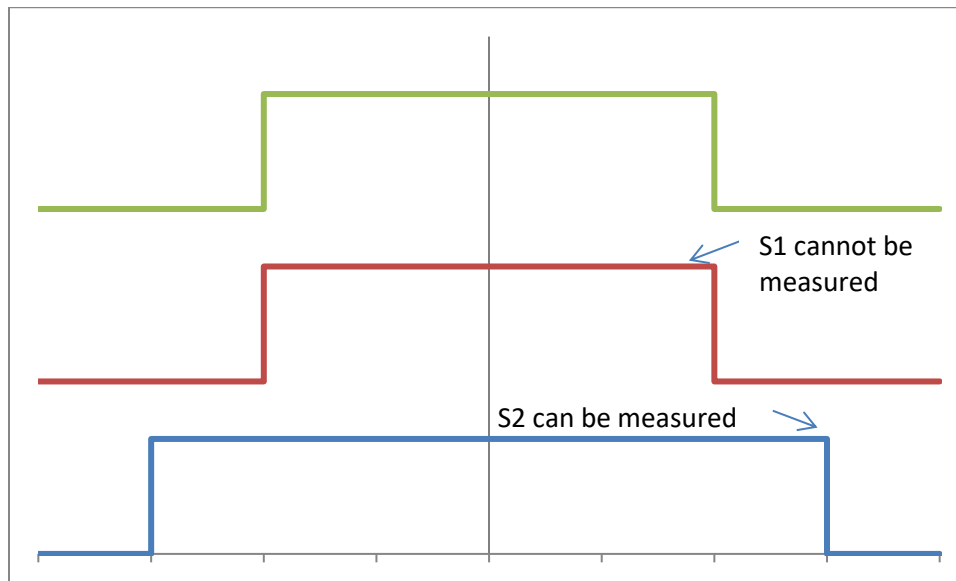


Figure 3: Case where one sampling point is not measureable

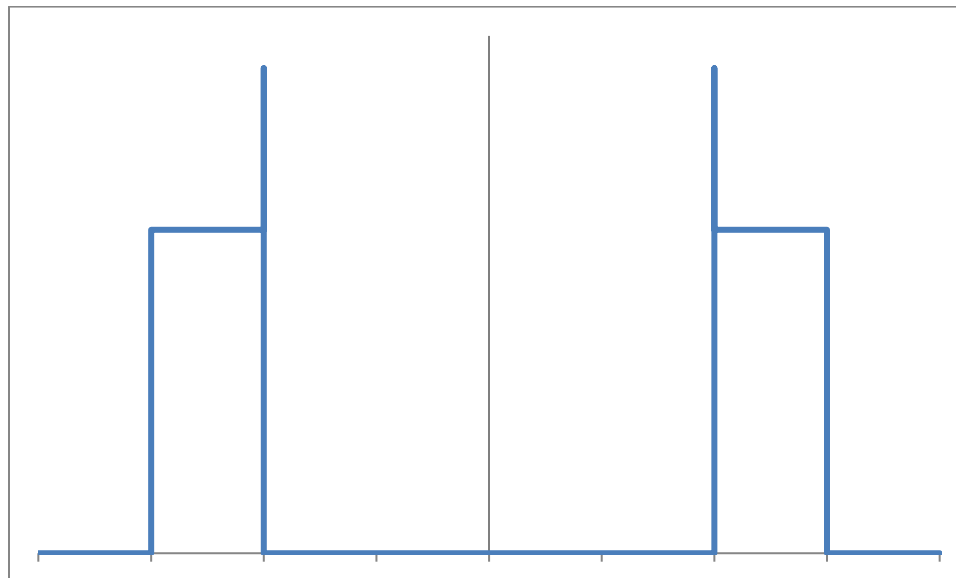


Figure 4: Resultant voltage waveform across shunt for Figure 3

In order to get around the issue of the small gap, the single-shunt algorithm modifies the duty cycles so that the minimum gap is maintained during that particular control cycle. This technique is shown in Figure 5 and Figure 6.

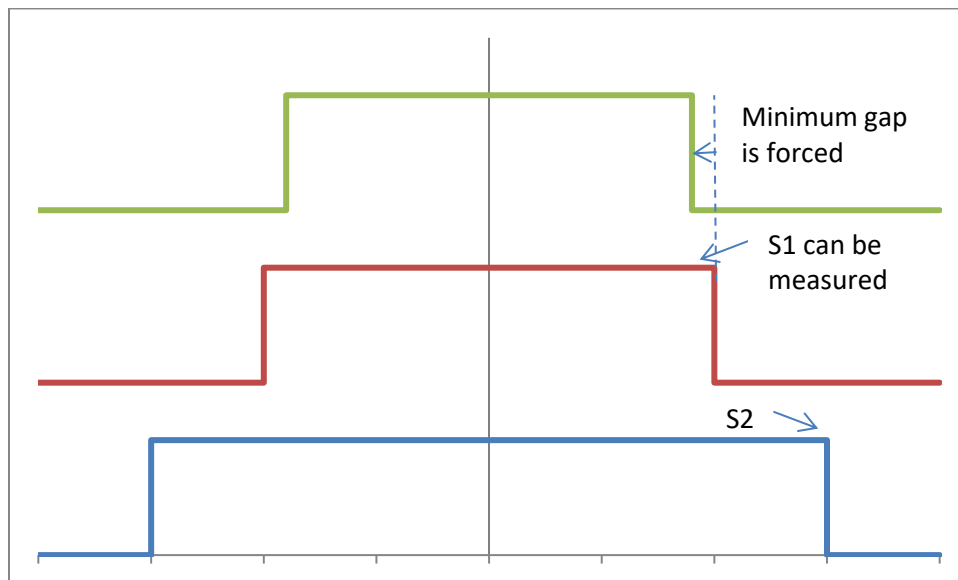


Figure 5: Case where minimum gap is forced

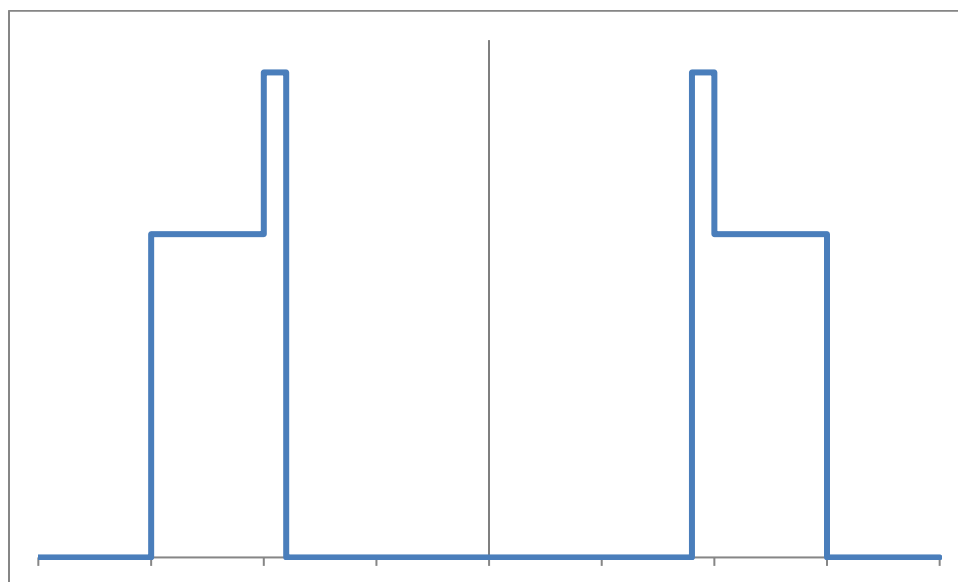


Figure 6: Resultant voltage waveform across shunt for Figure 5

2.3 Minimum Gap Compensation

When the minimum gap is forced, it generates a distortion in the applied voltage and therefore also results in phase current distortion. In order to reduce the effects of this distortion, the 1-shunt algorithm also implements a form of compensation.

3 FIRMWARE FLOW

The 1-shunt implementation consists of two main elements:

- 1) Phase current reconstruction from the 2 shunt current samples
- 2) Duty cycle manipulation to maintain the minimum gap required to measure the shunt current samples

The flow diagram from

Figure 7 illustrates a top-level view of the implementation.

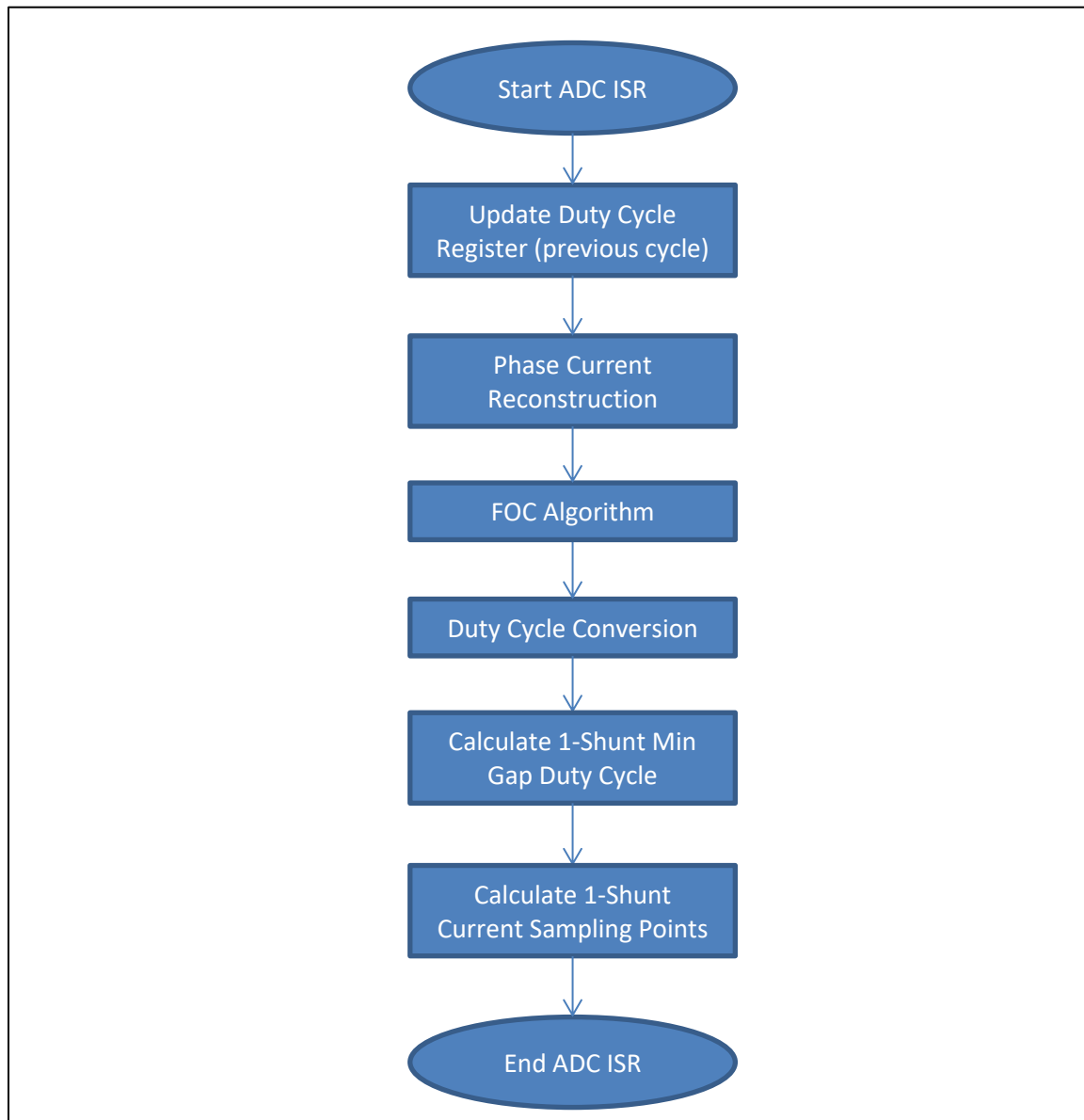


Figure 7: Single-Shunt Control Flow

4 HARDWARE CONFIGURATION

Sensing motor phase current in 1-shunt mode requires only one current shunt as shown in Figure 8.

Note that the filtering used here needs to have a very small RC time constant to reduce the effects of filtering on the sampling level.

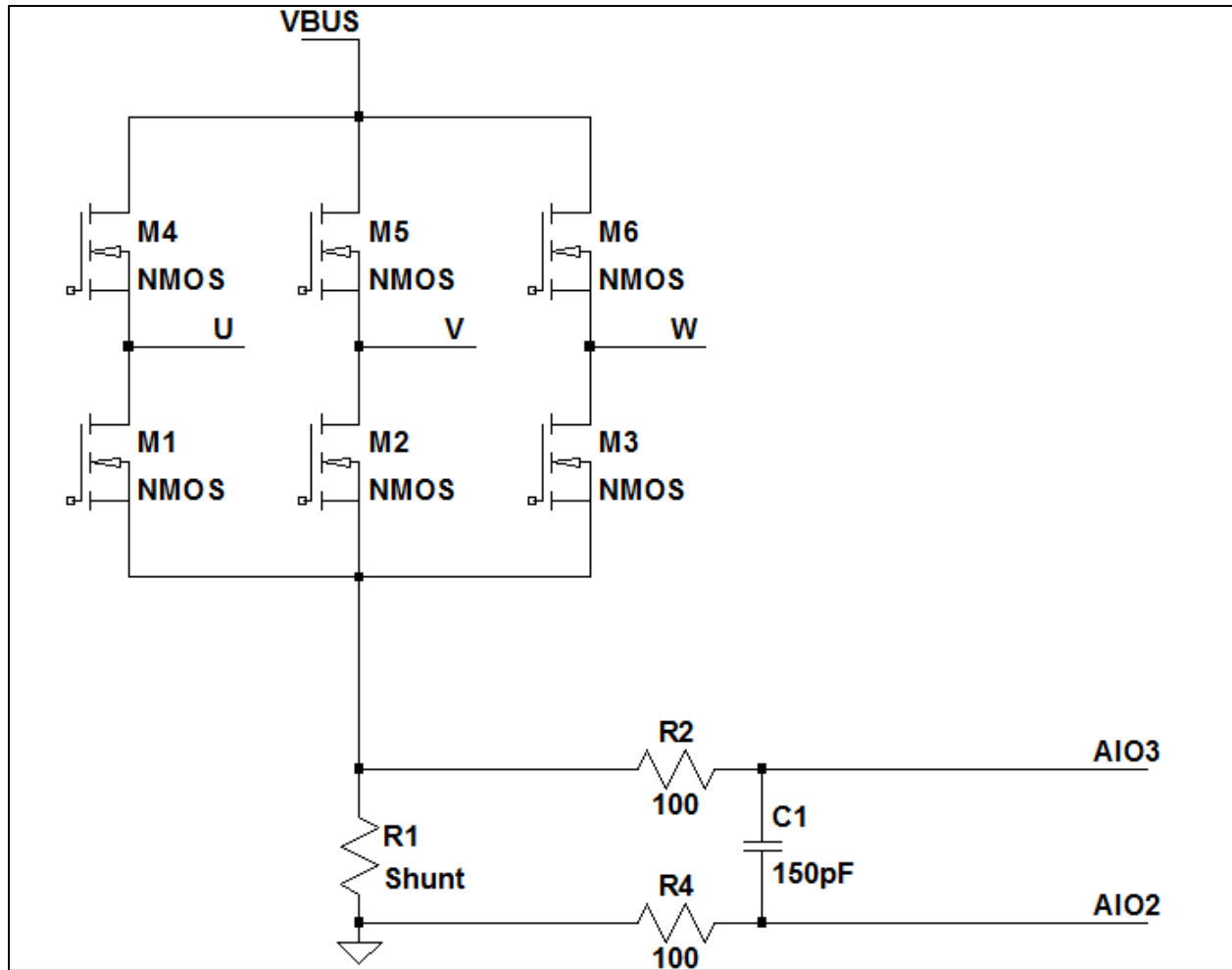


Figure 8: Single-Shunt Sensing Requirements

5 SOFTWARE CONFIGURATION

5.1 AIO Port Configuration

Only one diffamp needs to be configured for single shunt operation since all 3 phase currents are calculated from 2 measurements from the same shunt resistor. Overcurrent protection is also only necessary for the diffamp channel being used.

Below is a summary of the diffamp port settings required for a board with 1-shunt configuration:

- Number of Ports: 1
- Port Mode: DiffAmp
- Gain: Application dependent (set by GUI)
- Mid-Range Offset: Enabled
- Overcurrent Protection: Enabled on 1-shunt port only

5.2 Sequencer Configuration

The sequencers are configured to sample the 2 shunt voltages as well as the VDC bus voltage. This is currently done by using 2 sequencers, one for measuring the shunt current for the 1st edge and one for the shunt current from the 2nd edge and the bus voltage.

The details of the sequencer configuration are explained below:

- 1) Sequencer slot assignments:
 - a) S0 / N0 : Motor Phase Current 1
 - b) S1 / N0 : Motor Phase Current 2
 - c) S1 / N1 : Vbus
- 2) Sequencer Trigger Configuration:
 - a) Trigger Mode : S0 and S1 independent trigger
Triggers need to be independent because each sequencer will sample the shunt current at different times.
 - b) Sequencer Trigger CCR: S0=A0 & S1=A1
 - c) Sequencer Trigger Edge: High-to-Low
 - d) Sequencer Trigger Value:
Set dynamically to follow the appropriate PWM edges as shown in Figure 2.
- 3) ADC ISR Trigger: S1
The control loop runs after both samples from S0 and S1 are complete. S1 trigger is always set to trigger S1 after S0.



6 USING SINGLE-SHUNT MODE

There are no parameters to adjust in the GUI or in the firmware for single shunt operation.

To use the FOC project in single shunt mode, please follow the following steps:

- 1) Ensure that a single shunt board is selected in config_bpard.h and compile project.
 - a. Example: BOARD_EP5223LVM1_1_1SHUNT
- 2) Load GUI parameters and use the GUI in the same way as when working with the 3-shunt configuration.

Notes:

Note: Attempting to operate a 1-shunt board with a configuration file that is not 1-shunt may result in PCB damage!

7 SINGLE-SHUNT PARAMETERS

7.1 External Parameters

There are no parameters to adjust in the GUI or in the firmware for single shunt operation.

7.2 Internal Parameters

The following parameters are defined for single-shunt configuration. These are internal and cannot be modified without re-compiling the control library.

`S1_SAMPLE_ADVANCE_TICKS`

Number of TimerA tick counts that the sequencer will trigger ahead of the PWM edge

`S1_MINGAP`

The minimum gap in TimerA tick counts between 2 PWM edges that the single shunt algorithm considers acceptable for a good current measurement

Note: Changing these internal parameters without compiling the control library may result in improper operation!

8 LIMITATIONS OF 1-SHUNT OPERATION

Following is a list of performance limitations when operating in 1-shunt mode:

- 1) Potential issues with fast speed changes, presumably because when mingap is forced during a large acceleration the distortion may be too large to recover when few control periods are running compared to the electrical cycle.
- 2) Audible noise and increased current distortion while operating in the low speed range of the motor.
- 3) The higher the PWM frequency, the larger the % mingap out of period. Therefore in 1-shunt it is best to operate at the lowest possible PWM frequency that the application will allow.
- 4) Higher switchover speed may be necessary to avoid noise and distortion at low speeds in Closed Loop.
- 5) Cannot use coasting feature due to increased number of current sampling collisions. During coasting the voltage vector becomes very small and therefore it becomes more difficult to maintain the minimum gap.
- 6) Cannot go to 0% duty cycle because current cannot be measured, so only basic duty limit can be implemented, no band exclusion. As a result, the full speed potential of the motor may not be achieved.
- 7) Control frequency up to 27kHz is supported in 1-shunt mode and 30kHz in 3-shunt mode.



CONTACT INFORMATION

For the latest specifications, additional product information, worldwide sales and distribution locations:

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